Structure-Aware Graph Neural Networks for Binding Affinity Estimation

Lena Fischer, Haruto Sato
Preprint, 2024
Synthetic demo paper generated for ResearchMind AI testing. Not a real publication.

Abstract

Binding affinity estimation benefits from knowledge of the three-dimensional protein-ligand complex. We propose SA-GNN, a graph neural network that operates on 3D protein-ligand interaction graphs. We demonstrate that incorporating 3D structure improves affinity prediction. On the PDBbind dataset, SA-GNN reduces RMSE by 12% compared to sequence-based models.

1 Introduction

Sequence-based affinity models ignore the geometry of binding pockets. We argue that explicit 3D structure is necessary for accurate affinity estimation.

2 Method

SA-GNN builds an interaction graph from the protein-ligand complex, connecting atoms within 5 angstroms. Message passing is performed with a Graph Attention Network layer followed by an equivariant graph neural network layer.

3 Experiments

We train and evaluate on the PDBbind dataset core set and additionally test on BindingDB complexes with docked poses. Results show that SA-GNN achieves an RMSE of 1.21, a 12% reduction compared to DeepDTA, and a Pearson correlation of 0.83.

4 Limitations

SA-GNN requires experimentally resolved or docked 3D structures, which are unavailable for many targets. We did not evaluate on cold-start protein splits or on the DAVIS and KIBA benchmarks. Performance with predicted structures from AlphaFold remains untested.

5 Conclusion

Structure-aware message passing improves binding affinity estimation when 3D complexes are available.

References

[1] Placeholder reference for the synthetic paper.